

Lab 3: Helicopter Control Design

Analogue And Digital Implementation



EEE3094S Control Systems Engineering

MCBTIN001_PTLAKH001

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Introduction

This report documents an altitude controller design for the helicopter system model which was modelled as a first-order system using system identification methods in the previous lab. A systematic approach was used to implement the plant controller, from *specifications-design-simulation- implementation-testing*. The objective is drive the helicopter system to a desired altitude while minimizing under/overshoot, steady-state error and good set point tracking and some level of stability with the help of control theory.

Feedback Control Theory

A feedback control system is a system that monitors the plant output by taking sensor measurements and continuously compares it to the set point to maintain the desired system output. The difference (error signal) is processed by the controller, which then makes the right adjustments to keep the system output within the desired range thus acting as a compensator for the system.

Because of feedback control, the controller will be able to detect any changes in the system output and then produces a control signal as an input to the plant to achieve the desired output. A feedback system is used to overcome the limitations of open-loop controllers. The advantages of closed-loop controllers include:

- Improved set point tracking
- Stable system performance
- Disturbance rejection
- Decreased sensitivity to system parameter deviations

An example of closed-loop controller is the Lead-Lag Compensator. This report considers various controller designs to determine the best design to meet the client specifications with the best performance.

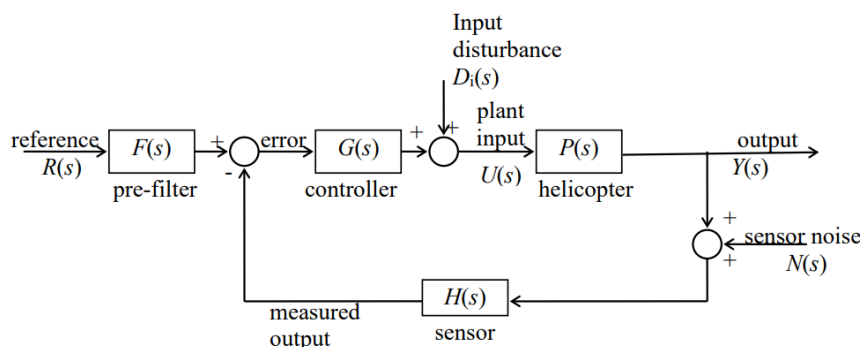


Figure 1: Helicopter Feedback System Components

Plant System Model

$$P(s) = \frac{13.6856}{12.33s + 1}$$

Client Specifications

- Set Point Tracking must be within 0.4m of the final value
- Over/undershoot must be within 20%
- Settling time (10s) must be within $t_{2\%}$
- Output disturbance rejection. Return to the previous final value must be within $t_{2\%}$

Designs and Simulations

Design 1: Lead-Lag Controller

Figure 2 shows the design of the lead-lag controller, $G(s)$ using sisotool in MATLAB. This controller was designed by moving the pole and the zero to different locations in the root locus to meet the desired specifications. The overshoot is within 20% and setting time of **9.95s**. The set point tracking is within the 0.4m of the final value.

$$G(s) = \frac{14.254(s + 0.1395)}{s + 14.77}$$

Figure 2: Sisotool for the lead-Lag compensator

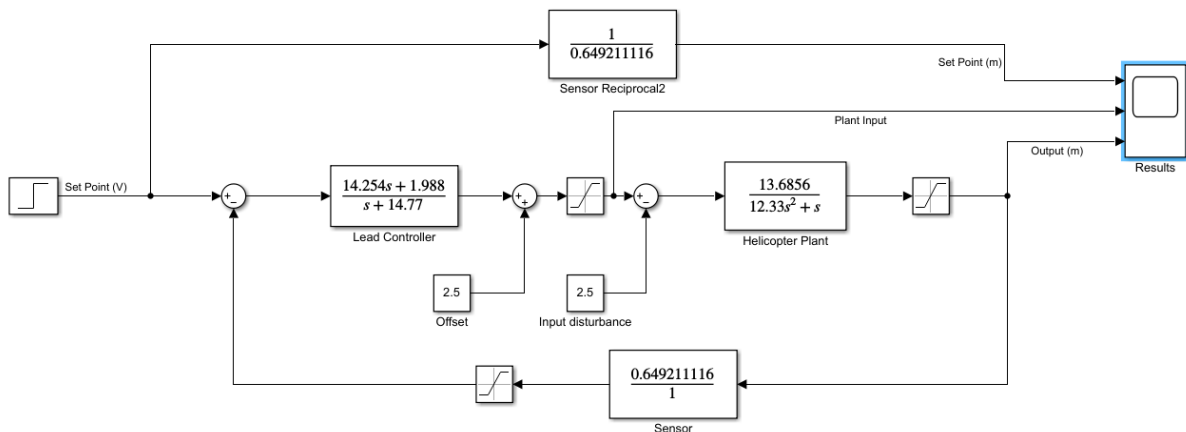


Figure 3: Lead-Lag Compensator Simulink Simulation

Figure 4 shows the performance of the lead-Lag controller from simulink simulation

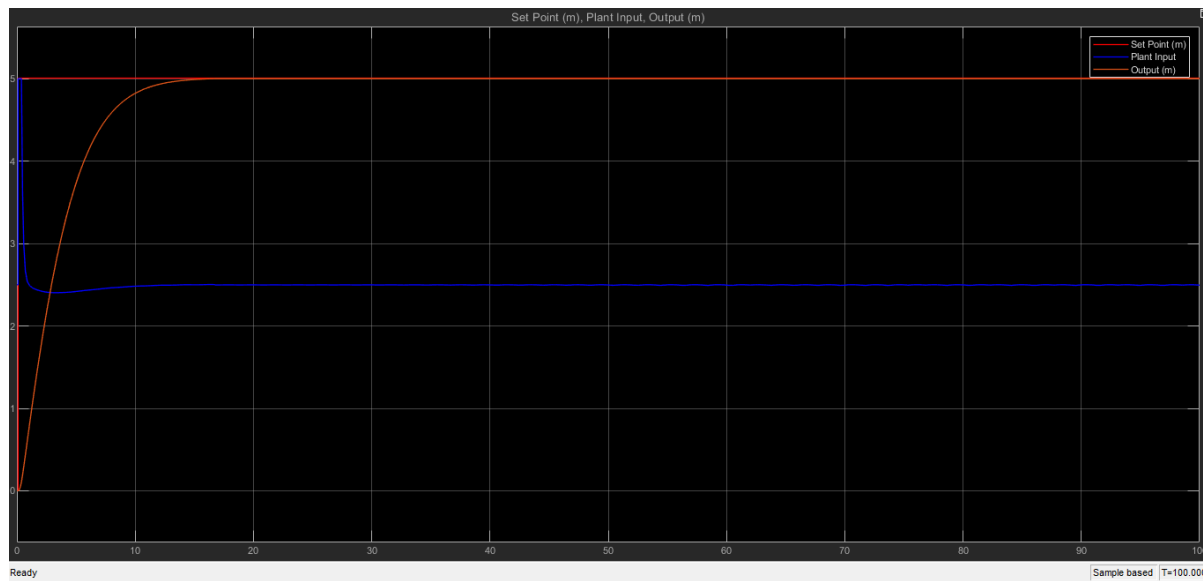


Figure 4: Scope results of the lead-lag controller

Design 2: Lead Controller

The lead controller was designed by placing one lead pole and one zero on the root locus using sisotool in MATLAB. The achieved overshoot is **3.21%** and settling time of **9.15s**. The lead controller shows a good performance, but it doesn't meet all the specifications and it has a slightly higher steady state error.

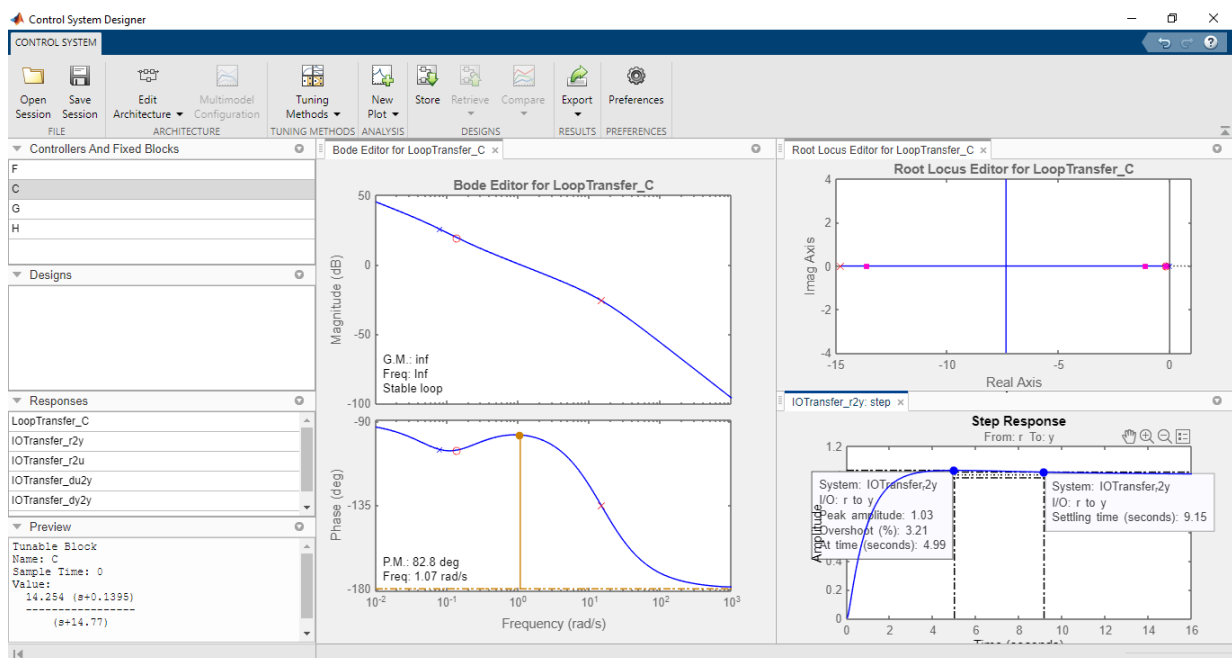
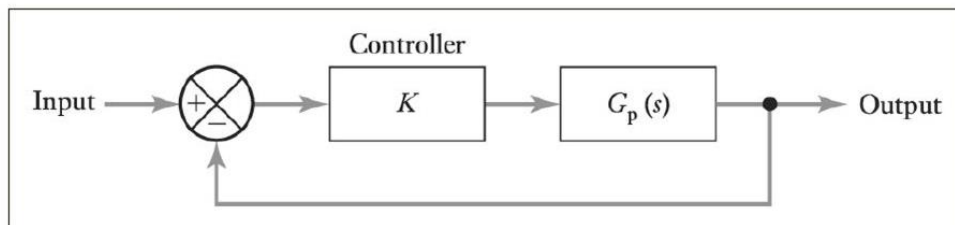


Figure 6: Sisotool of the Lead Controller

Design 3: K Controller

Gain controller was included in the design because it describes the behaviour of the plant system in response to varying dynamics. K controller determines how far to adjust the controller output when the system variable deviates from the set point. The gain controller analogue implementation only requires the gain amplifier component and its one of the easiest designs.



Comparison and Final Design

1. Lead-Lag Controller: The first design meets all the clients specifications. It has the smallest steady state error compared to the other designs. It has a settling time is closer to the expected setting time and within $t_{2\%}$. It is able to track the set point fairly well. The overshoot is within 20%.
2. Lead Controller: The lead controller shows a good performance as well, but it does not meet the specified overshoot and the steady state error is much higher compared to the lead-lag compensator.
3. Gain Controller: The plant is able to stabilise but it oscillates a lot. The system is only able to track the setpoint but all other specifications are not met.

The chosen final design is the **lead-lag controller** shown below which meets all the specifications.

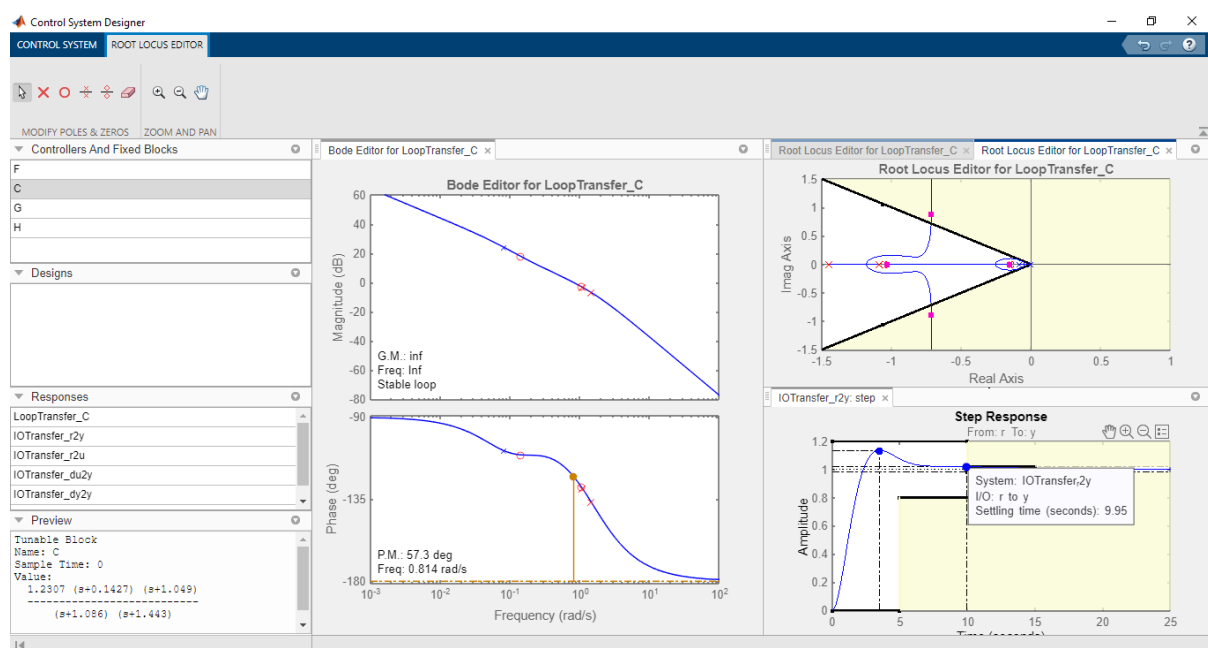
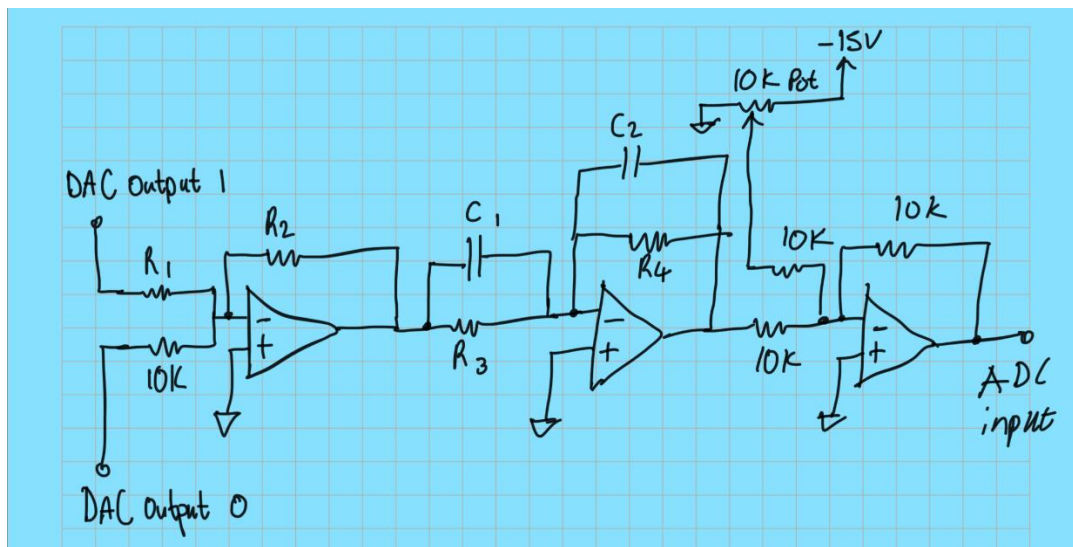


Figure 7: Sisotool of the lead-lag Controller

Analogue Circuit Design and Implementation in Practice

The fig below shows the circuit diagram for the lead-lag controller analogue implementation. It uses 3 UA741cp op-amps, where two are used for summing and the other one for the controller. The UA741 was chosen because it has good slew rate and it can handle maximum supply voltages of -18V to +18V whereby the lab station supply -15V to +15V.

$$G(s) = \frac{R_2 R_4 (R_1 C_1 s + 1)}{R_1 R_3 (R_2 C_2 s + 1)}$$



$$G(s) = \frac{14,254 (s + 0,1395)}{(s + 14,77)} = K \left[\frac{1 + 7,168s}{1 + 0,067s} \right]$$

Standard E12 Series Resistors and Capacitors Were Used

$$\frac{R_2}{R_1} = 14,254, \text{ let } R_1 = 10\text{k}\Omega, R_2 = 142,54\text{ k}\Omega \\ R_2 \approx 150\text{ k}\Omega$$

$$a = 7,168 = R_3 C_1, C_1 = 33\mu\text{f}, R_3 = 215\text{k}\Omega$$

$$b = 0,067 = R_4 C_2, C_2 = 33\mu\text{f}, R_4 = 2\text{k}\Omega$$

Testing

Laboratory results for continuous time design

The figure below shows the response of our analogue controller circuit with all specifications met. The overshoot is within 20% and the settling time of 10s. It also shows a good disturbance rejection. The plant is able to track the set point within the specified range as well as sinusoidal tracking. However, as the amplitude of the sinusoid increases, it starts to fall a little bit behind but the overall response is fairly well.

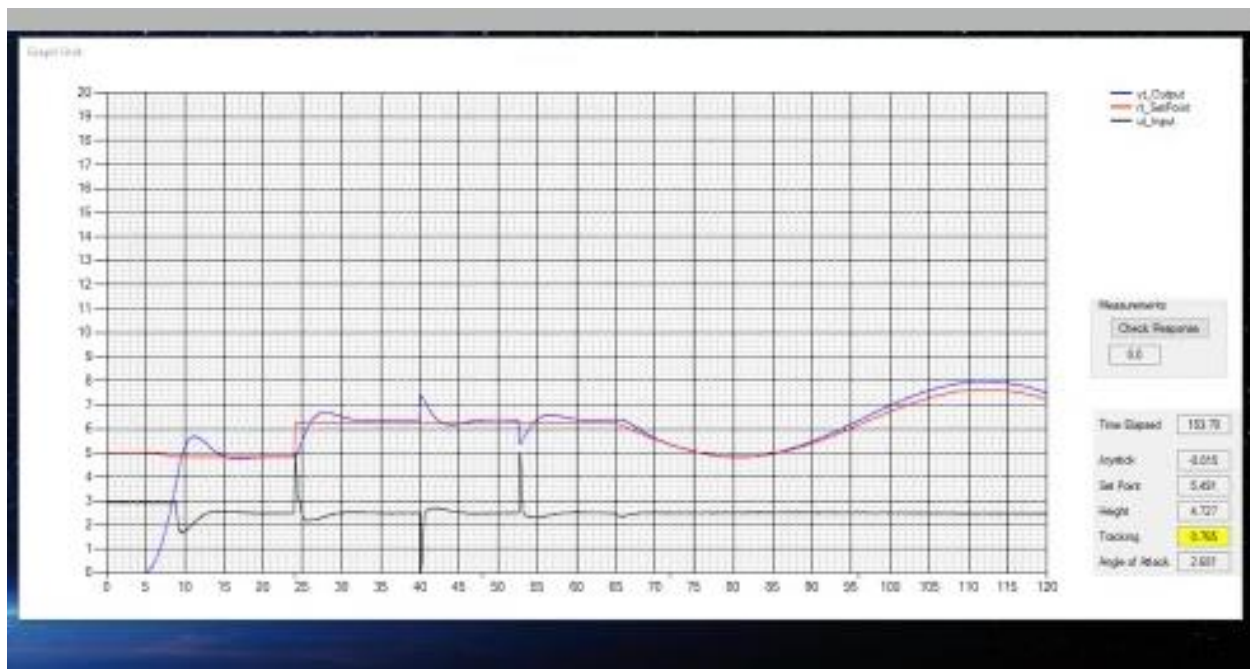


Figure 8: Testing of the analogue implementation of our controller by the client , Prof De Mar

Conclusion

The results from figure 8 shows that the controller design met all the specifications. The analogue circuit design took a lot of time to be successfully implemented compared to using MATLAB tools. This was bought about having to choose the right resistor values and applying the right electronics knowledge, not only control theory. The analogue circuit response looks similar to the simulations of our lead-lag controller.